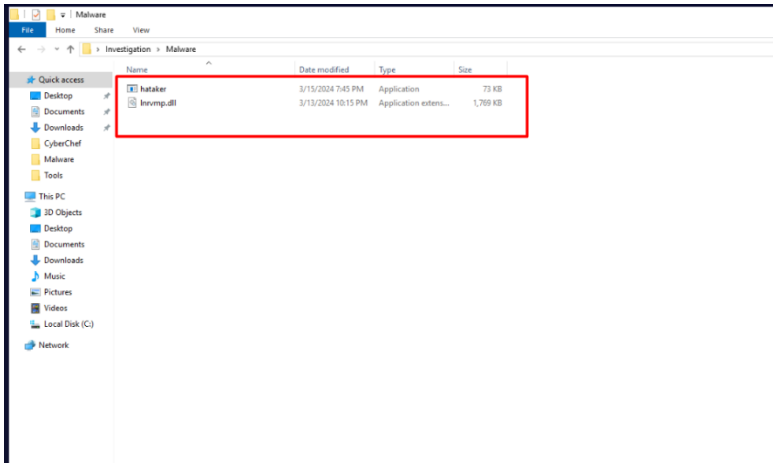
Reverse Engineering.

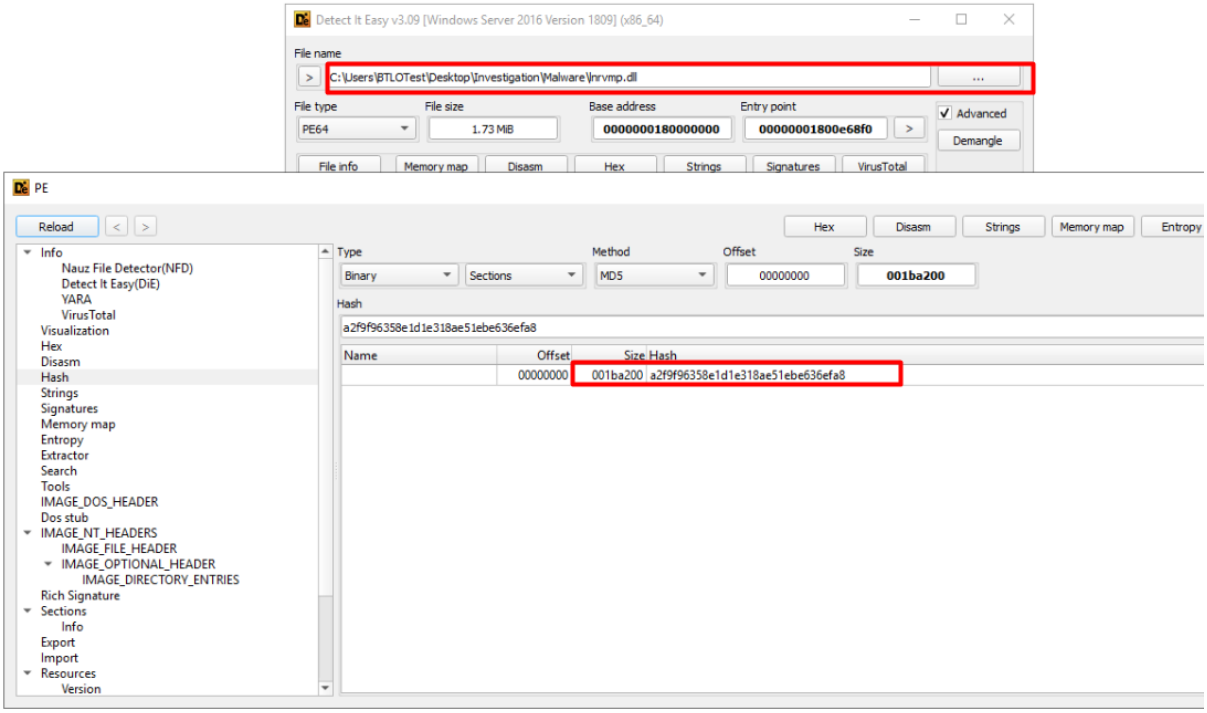
Investigation Reference: https://blueteamlabs.online/home/investigation/anakus-dfea6f86e0

Tools Used: Detect It Easy, Virustotal, Timeline Viewer, Wireshark, Powershell, Sysinternals suite

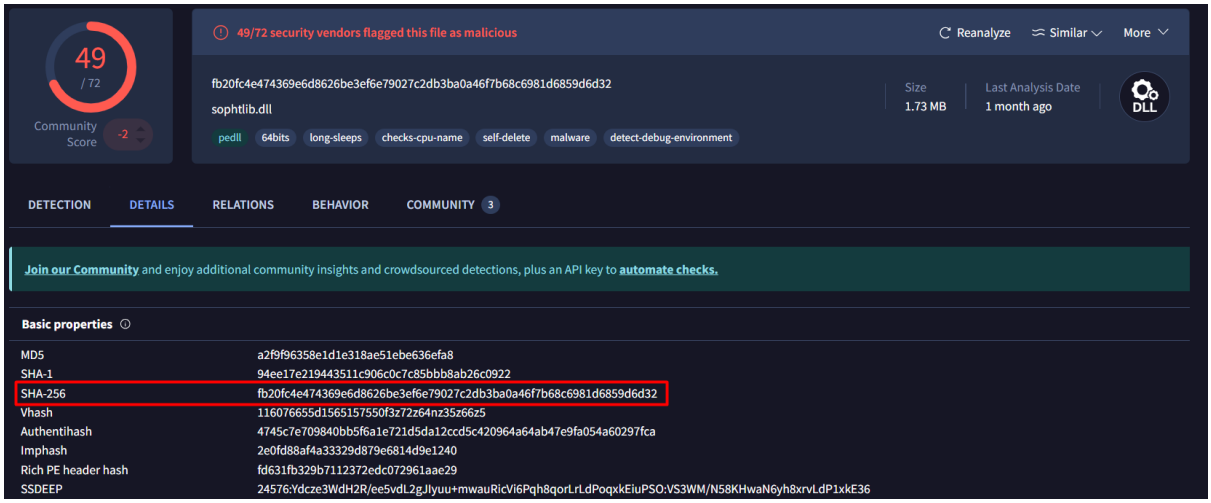
Two malware files have given in this investigation.

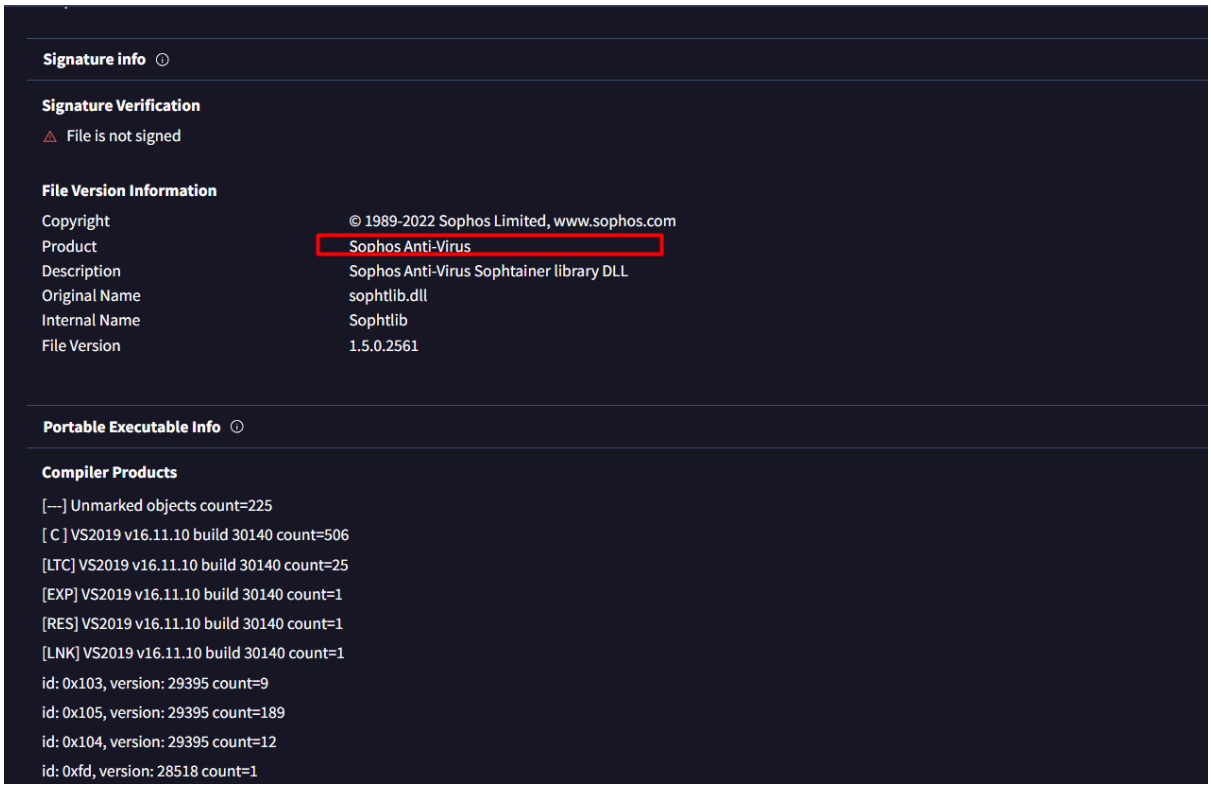


Generate the sash File of lnrvmp.dll and check its reputation in Virustotal.



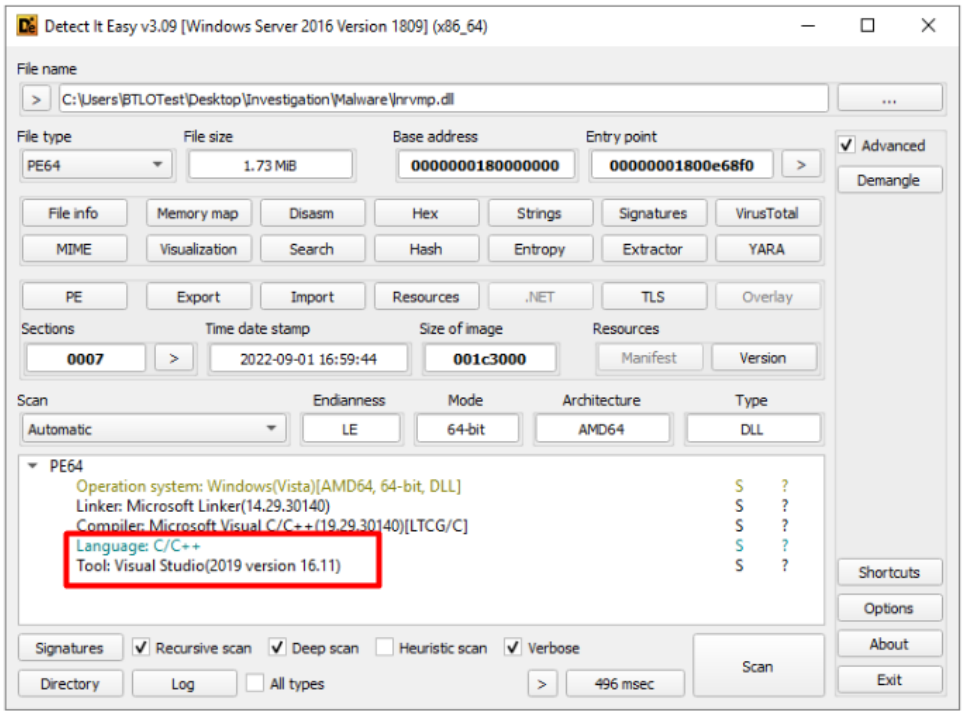
Reputation check in Virustotal.



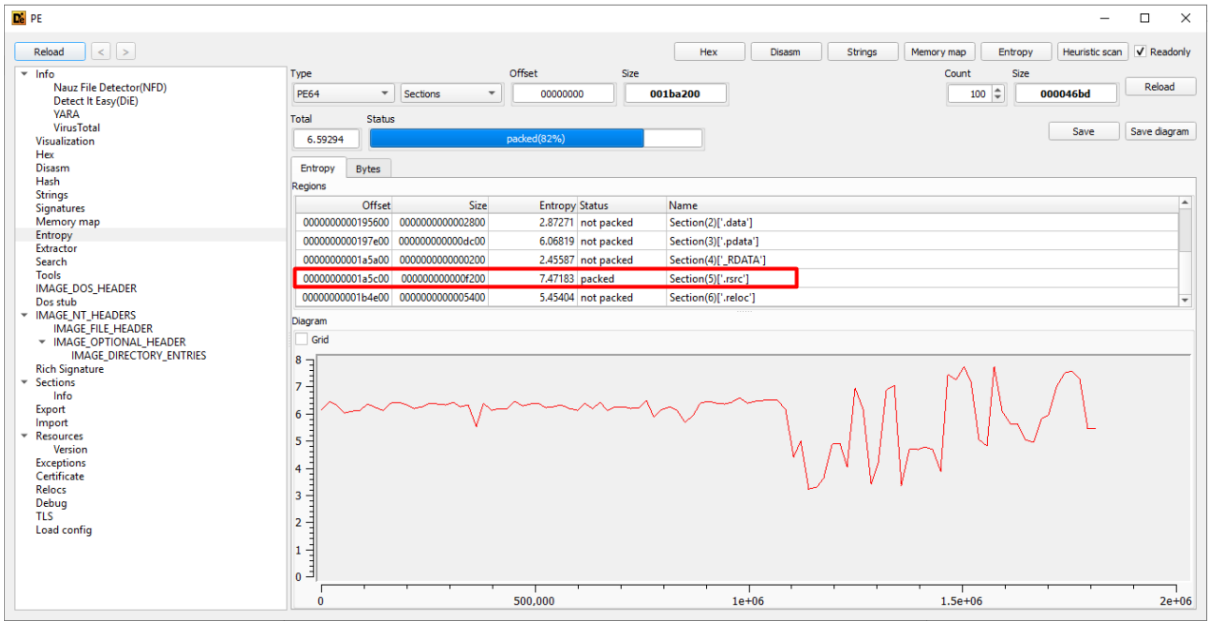


Malware is impersonating Sophos Anti-Virus. Virustotal indicates it as a malicious.

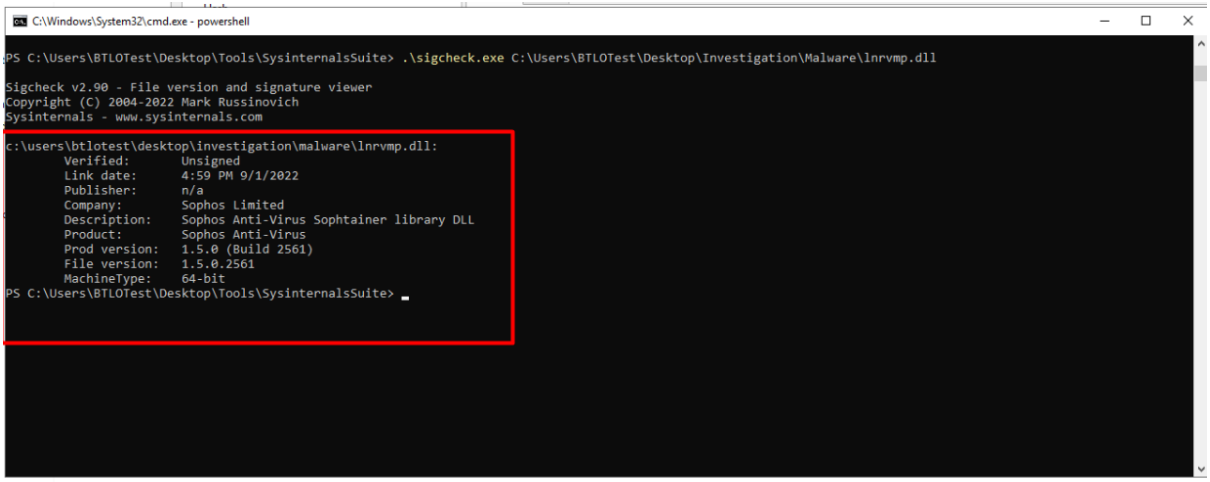
Programming language used to write it:



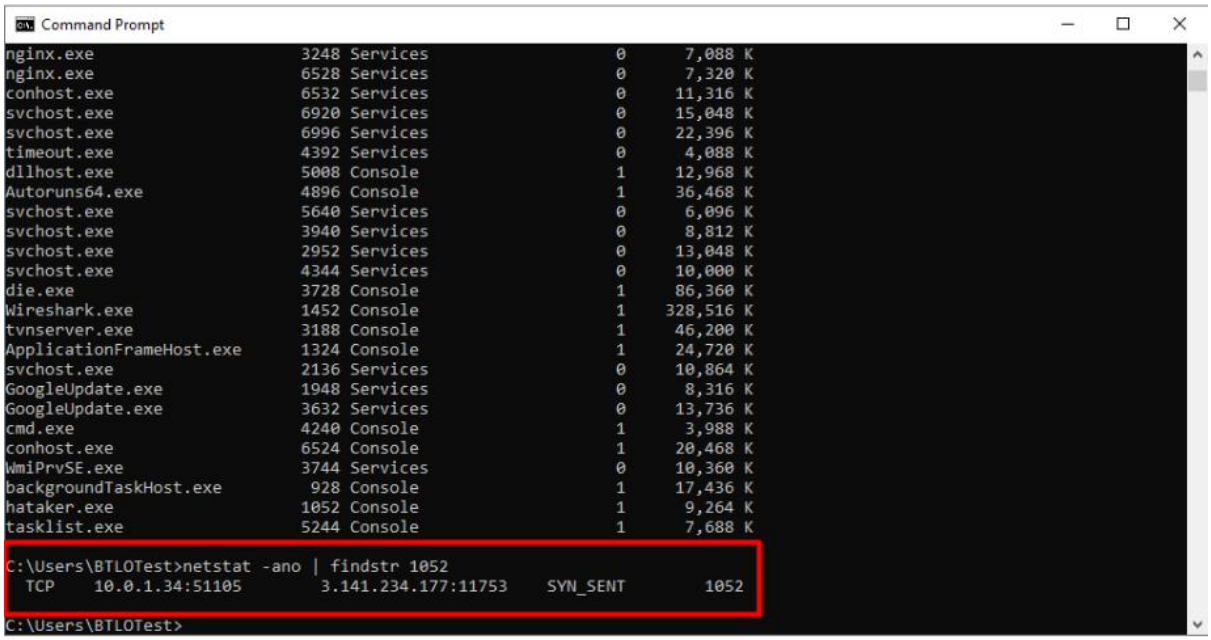
The total entropy value in Detect it Easy gives us a general indication of the randomness across the entire file, but the presence of a highly entropic-packed section indicates a portion of the file containing data that has been compressed—packed. Usually, an entropy above 7.2 is considered malicious



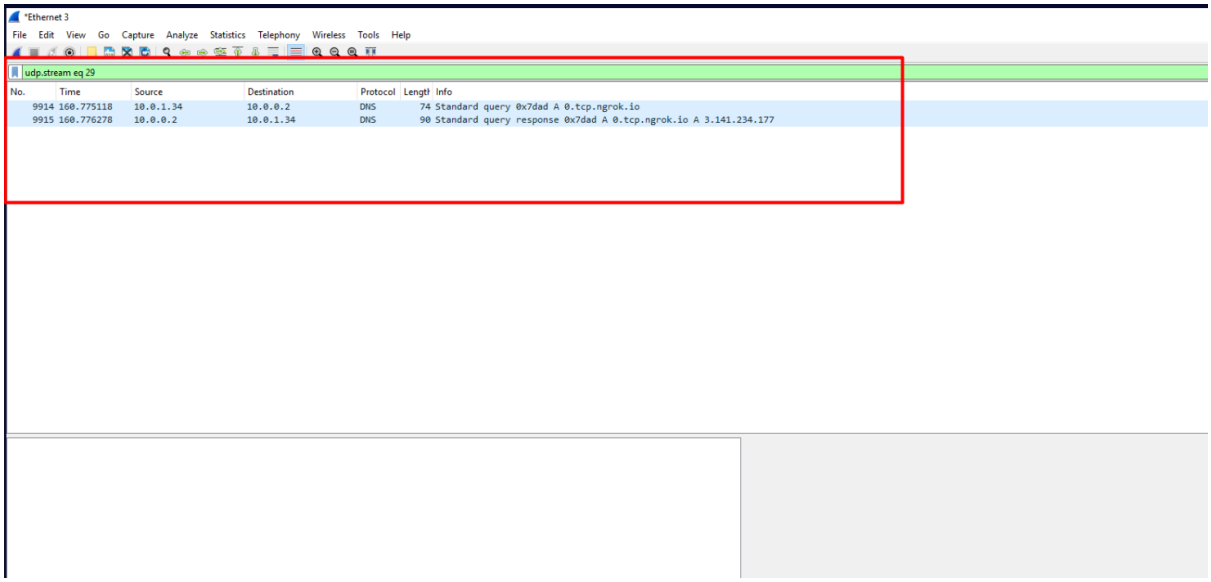
Checking with sysinternal to verify it is signed or not.



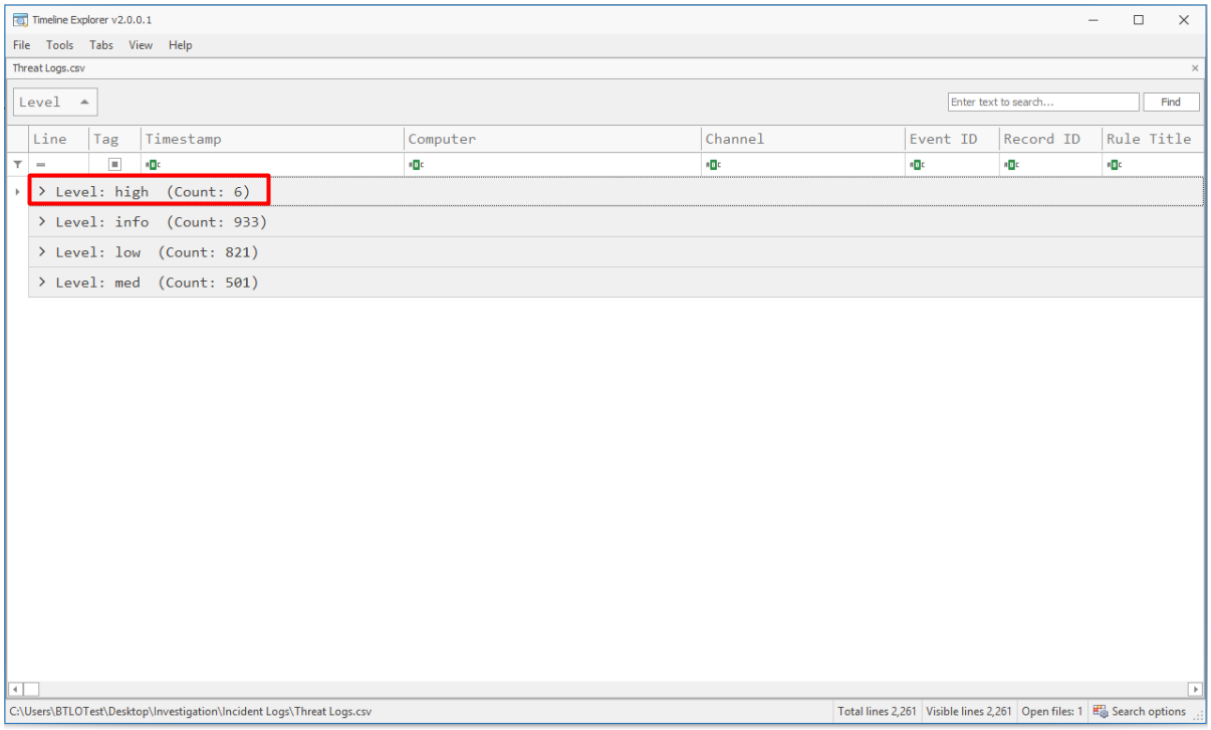
Hatker.exe is running in PID 1052

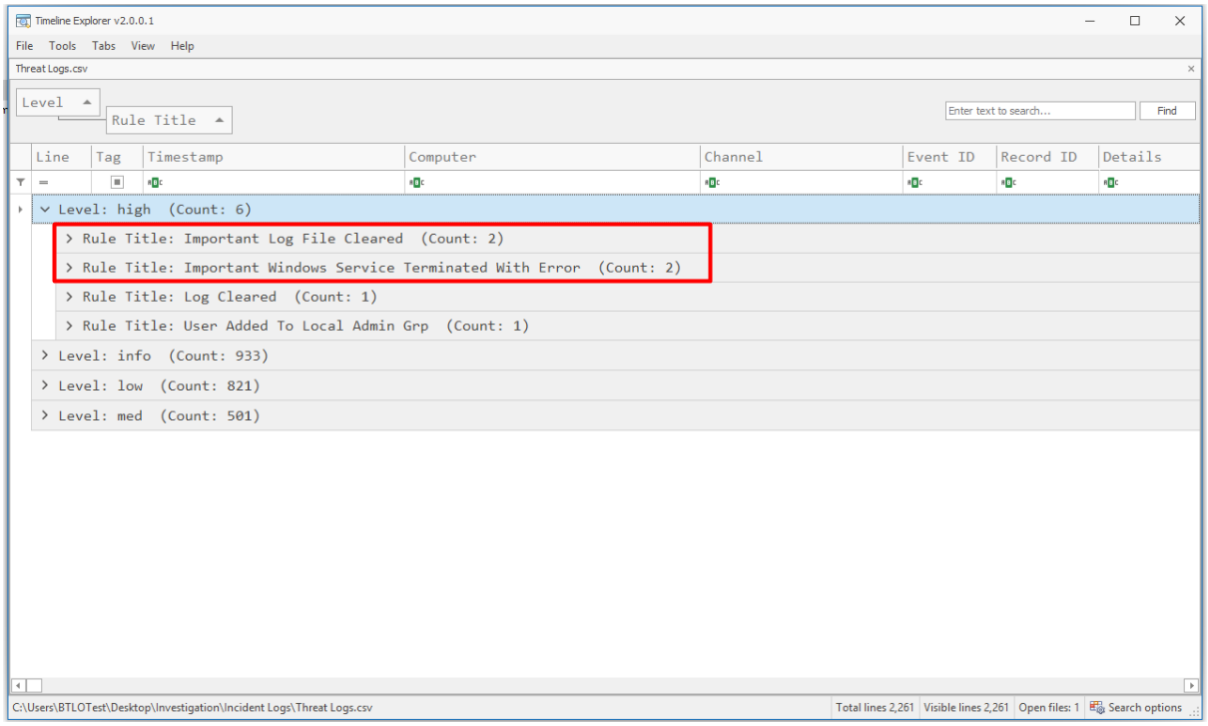


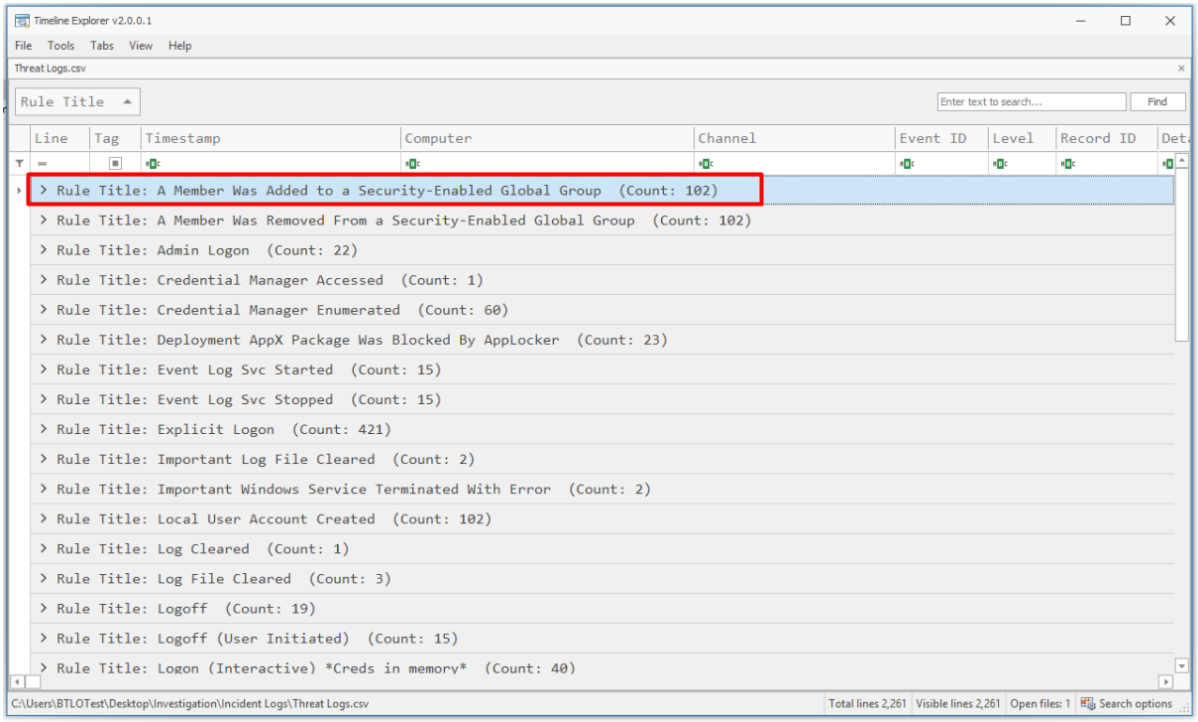
In the wireshark if we capture the network, we can see that unusual dns name as below.

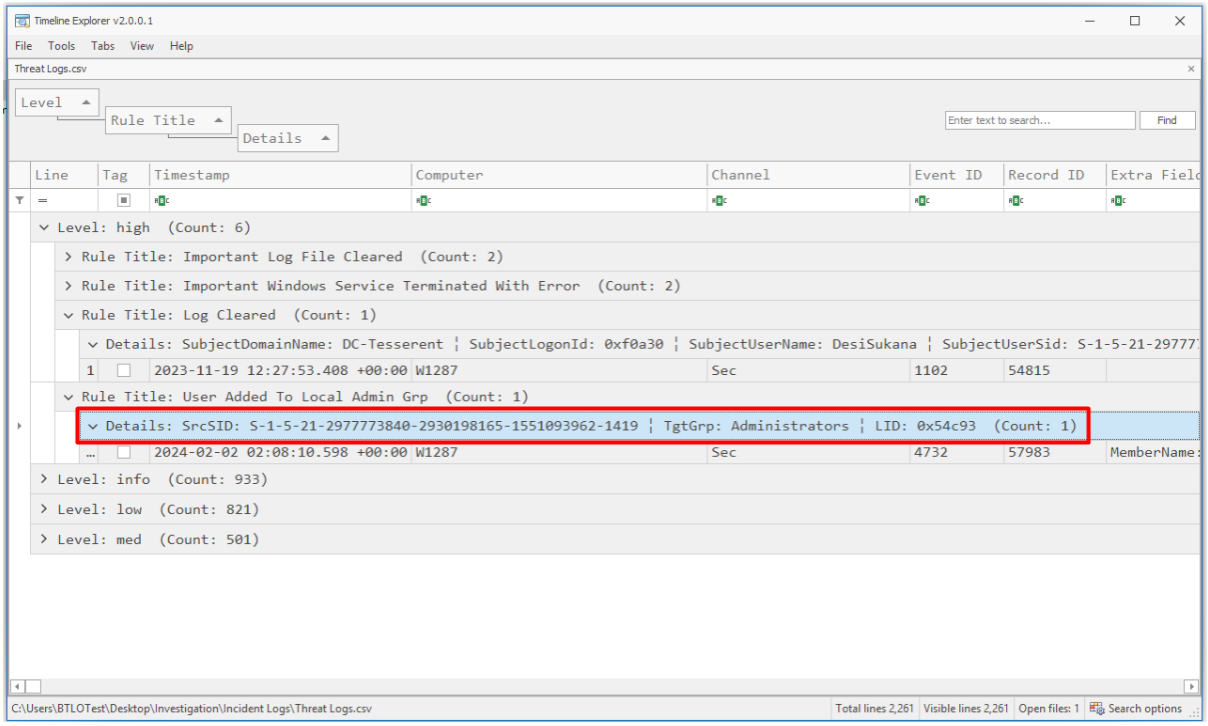


Timeline Explorer, look at the “Threat Logs” from the incident, how many high-risk alerts were tracked




Timeline Explorer v2.0.0.1
File Tools Tabs View Help
Threat Logs.csv
Level
Rule Title
Enter text to search...
Find
Line Tag Timestamp Computer Channel Event ID Record ID Details
Level: high (Count: 6)
Rule Title: Important Log File Cleared (Count: 2)
Rule Title: Important Windows Service Terminated With Error (Count: 2)
Rule Title: Log Cleared (Count: 1)
Rule Title: User Added To Local Admin Grp (Count: 1)
Level: info (Count: 933)
Level: low (Count: 821)
Level: med (Count: 501)
C:\Users\BTLOTest\Desktop\Investigation\Incident Logs\Threat Logs.csv
Total lines 2,261
Visible lines 2,261
Open files: 1
Search options


Timeline Explorer v2.0.0.1
File Tools Tabs View Help
Threat Logs.csv
Rule Title
Enter text to search...
Find
Line Tag Timestamp Computer Channel Event ID Level Record ID
Rule Title: A Member Was Added to a Security-Enabled Global Group (Count: 102)
Rule Title: A Member Was Removed From a Security-Enabled Global Group (Count: 102)
Rule Title: Admin Logon (Count: 22)
Rule Title: Credential Manager Accessed (Count: 1)
Rule Title: Credential Manager Enumerated (Count: 60)
Rule Title: Deployment AppX Package Was Blocked By AppLocker (Count: 23)
Rule Title: Event Log Svc Started (Count: 15)
Rule Title: Event Log Svc Stopped (Count: 15)
Rule Title: Explicit Logon (Count: 421)
Rule Title: Important Log File Cleared (Count: 2)
Rule Title: Important Windows Service Terminated With Error (Count: 2)
Rule Title: Local User Account Created (Count: 102)
Rule Title: Log Cleared (Count: 1)
Rule Title: Log File Cleared (Count: 3)
Rule Title: Logoff (Count: 19)
Rule Title: Logoff (User Initiated) (Count: 15)
Rule Title: Logon (Interactive) *Creds in memory* (Count: 40)
C:\Users\BTLOTest\Desktop\Investigation\Incident Logs\Threat Logs.csv
Total lines 2,261
Visible lines 2,261
Open files: 1
Search options



MITRE Overview.

Home > Techniques > Enterprise > Account Manipulation > Additional Local or Domain Groups

Account Manipulation: Additional Local or Domain Groups

Other sub-techniques of Account Manipulation (7)

An adversary may add additional local or domain groups to an adversary-controlled account to maintain persistent access to a system or domain.

On Windows, accounts may use the `net localgroup` and `net group` commands to add existing users to local and domain groups.[1][2] On Linux, adversaries may use the `usermod` command for the same purpose.[3]

For example, accounts may be added to the local administrators group on Windows devices to maintain elevated privileges. They may also be added to the Remote Desktop Users group, which allows them to leverage Remote Desktop Protocol to log into the endpoints in the future.[4] On Linux, accounts may be added to the sudoers group, allowing them to persistently leverage Sudo and Sudo Caching for elevated privileges.

In Windows environments, machine accounts may also be added to domain groups. This allows the local SYSTEM account to gain privileges on the domain.[5]

ID: T1098.007

Sub-technique of: T1098

Tactics: Persistence, Privilege Escalation

Platforms: Linux, Windows, macOS

Contributors: Madhukar Raina (Senior Security Researcher - Hack The Box, UK)

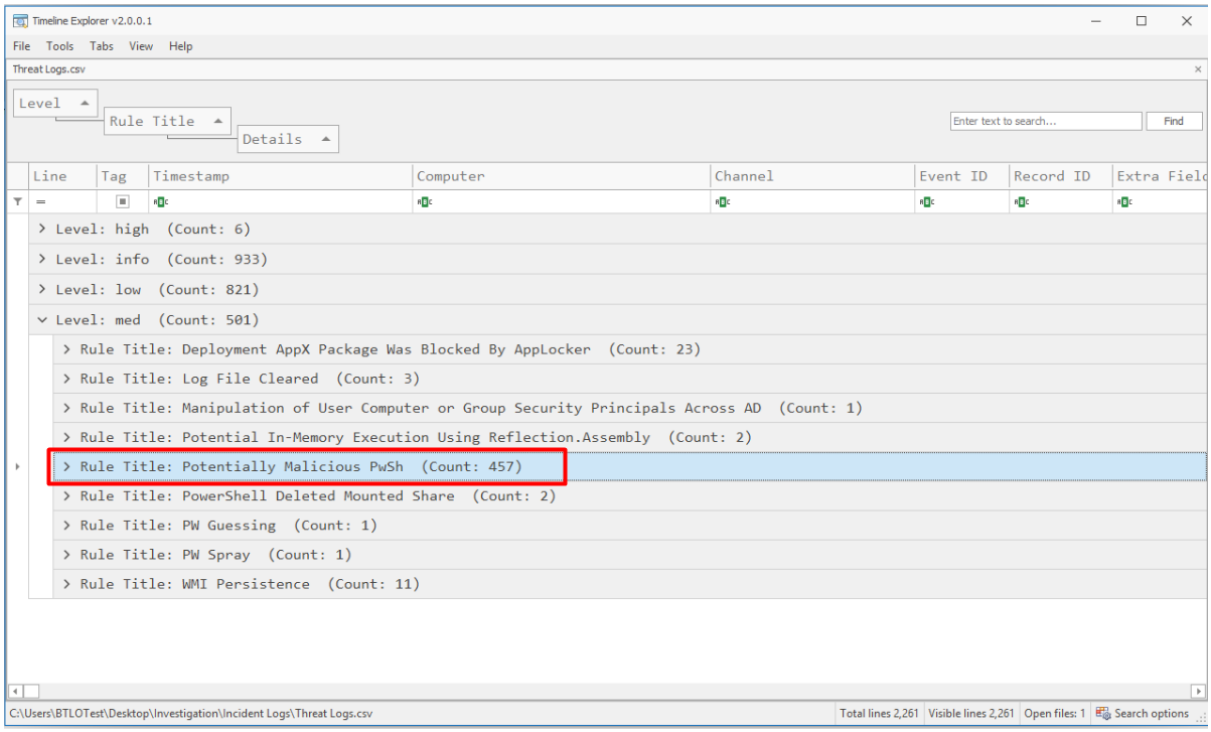
Version: 1.0

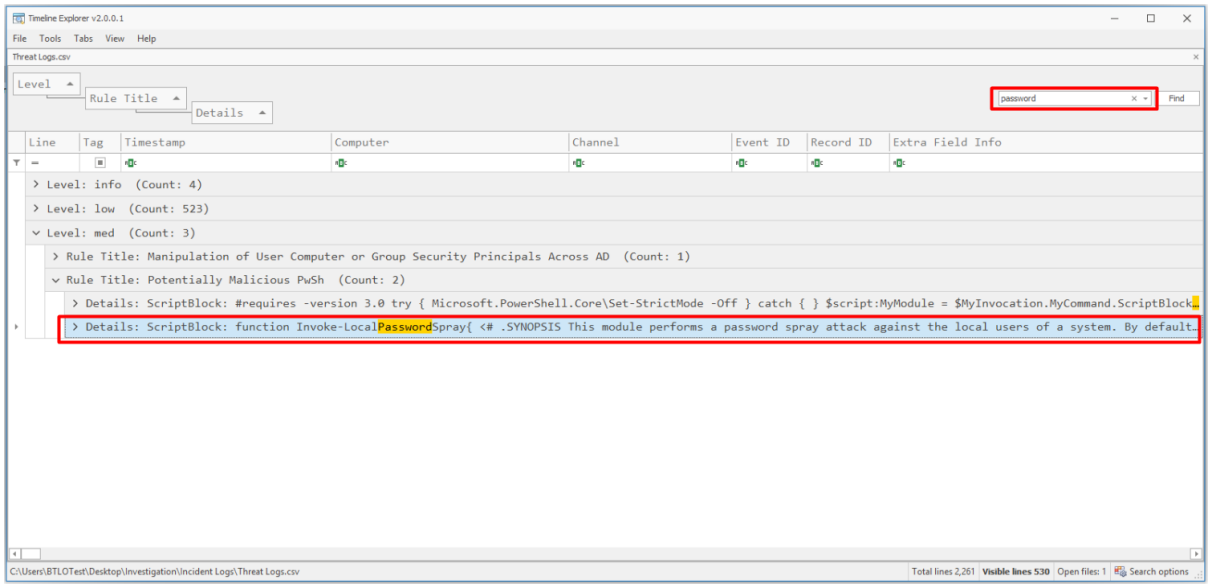
Created: 05 August 2024

Last Modified: 15 April 2025

Version Permalink

Procedure Examples





Takeaway:

1. Use of DIE tool to find about the file information.
2. Entropy means how random or unstructured the data inside a file is. Low entropy means data is very organized and predictable, not packed or encrypted. High entropy, closes to 8 means that file is packed, compressed, encrypted to hide its real contents.
3. Use of timeline explorer to read the csv log.
4. Mapping to MITRE.